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Validity and reliability analysis of the Turkish life satisfaction scale developed through artificial intelligence

Servet Atik¹ and Nuri Erdemir^{2,3*}

Abstract

This study evaluates the validity and reliability of a Turkish Life Satisfaction Scale developed using artificial intelligence (ChatGPT) to explore AI's potential in creating psychometric tools. The scale was tested on three independent samples of Turkish university students: 503 for Exploratory Factor Analysis (EFA), 301 for Confirmatory Factor Analysis (CFA), and 79 for test–retest reliability. EFA identified a unidimensional structure, accounting for 67.50% of the total variance (factor loadings .75–.89). CFA confirmed adequate model-data fit (e.g., $\chi^2/\text{sd} = 2.63$, RMSEA = 0.07). The scale demonstrated high internal consistency (Cronbach's $\alpha = .88$) and temporal stability (test–retest correlation = .95). Criterion validity was supported by strong positive correlations with established Life Satisfaction ($r = .74$) and General Well-Being ($r = .63$) scales. These findings indicate that AI can expedite scale development while yielding robust psychometric instruments. This research underscores the innovative potential of AI-supported psychometric tools in the social sciences and offers valuable insights for future scale development.

Keywords Artificial intelligence, Life satisfaction, Validity, Reliability, Scale development, University students

Introduction

Life satisfaction, a cornerstone of subjective well-being, represents an individual's overall contentment with their life, shaped by psychological, social, and environmental factors [1]. Research highlights its role as a critical indicator of mental and emotional health, with links to self-efficacy, stress management, happiness, and social relationships [2]. Traditionally, developing scales to measure such constructs has been resource-intensive and time-consuming. However, artificial intelligence (AI) offers a promising avenue for creating innovative and efficient psychometric tools. This study focuses on life

satisfaction, a widely studied concept in Turkish [3–7] and international literature [8–10], due to its prominence in positive psychology. By leveraging AI, specifically ChatGPT, this research aims to develop a valid and reliable life satisfaction scale, streamlining the scale development process. Life satisfaction's well-defined framework makes it an ideal construct for testing the accuracy and practicality of AI-generated scales. Ultimately, this study seeks to measure the psychological well-being and life satisfaction of individuals affected by recent challenges, such as pandemics and earthquakes, while assessing the strengths and limitations of AI in psychometric tool development.

Artificial intelligence

Artificial intelligence (AI) encompasses the development of systems that mimic human intelligence, including learning and decision-making, which are essential for automating complex tasks like scale development. Since

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2010, AI research in education and psychology has grown significantly, as evidenced by increased publications in Web of Science and Google Scholar [11]. AI holds transformative potential for psychological research by facilitating the rapid and efficient creation of measurement tools. This study employs ChatGPT, an advanced natural language processing (NLP) model developed by OpenAI, to generate a life satisfaction scale. Launched on November 30, 2022, ChatGPT uses extensive training data to produce coherent, contextually relevant text, making it well-suited for crafting scale items [12]. Preliminary studies indicate that ChatGPT can generate well-structured survey items, supporting our objective of developing a psychometrically robust scale [13].

In this study, ChatGPT's application extends to generating initial scale items, which are then refined through expert validation to ensure validity and reliability. This approach offers opportunities to enhance scale development efficiency and reduce resource demands, supporting psychological research and practice. However, it also presents challenges, including ethical concerns about AI-generated content (e.g., originality and bias), dependency on technology, and the need to verify the accuracy of AI outputs against established psychological standards. For instance, ensuring that AI-generated items align with theoretical constructs like life satisfaction requires rigorous human oversight, while over-reliance on AI could compromise researchers' methodological skills. Addressing these challenges necessitates clear guidelines for AI use in research, robust validation processes, and ongoing evaluation of its outputs [14]. By exploring ChatGPT's role in this process, this study contributes to understanding how AI can support psychological measurement while highlighting areas for improvement.

Life satisfaction

Life satisfaction, first introduced by Neugarten, Havighurst, and Tobin [15], is a well-established construct in psychological research, reflecting an individual's cognitive and affective evaluation of their overall quality of life [8]. It is shaped by living conditions and cultural contexts [16]. Diener [1] described life satisfaction as a cognitive appraisal process, underscoring the role of psychological factors. Beyond a general assessment, it includes dimensions such as health, relationships, work, and personal fulfillment [2].

The need for a new life satisfaction scale arises from identified gaps in existing measures, particularly regarding cultural adaptability, dimensionality, and validity across diverse populations [17]. While current scales provide foundational assessments, limitations in their applicability—such as a lack of dynamic

responsiveness to contextual factors—highlight the necessity for improved measurement tools.

Artificial intelligence (AI) technologies can play a transformative role in addressing these gaps by enhancing the scale development process. Utilizing AI in the life satisfaction scale development process not only exemplifies innovation in psychometric methods but also offers potential improvements in efficiency and precision. If reliable and valid results are obtained, AI can significantly accelerate scale development while ensuring robust measurement properties. However, the primary focus remains on advancing life satisfaction measurement rather than solely demonstrating AI's application [18].

Validity and reliability

Developing scales to measure psychological constructs is a complex and resource-intensive process, requiring significant time, cost, and human effort [19]. Using AI-based language models like ChatGPT to generate scale items represents a novel approach, with limited research exploring its application. For instance [20] employed ChatGPT in scale development, integrating expert opinions and literature reviews with AI-generated insights to create scale items. During this process, ChatGPT provided suggestions for adapting certain items from existing scales to multimedia learning environments, which were carefully evaluated to enhance the item pool. In another study, [21] examined the potential of ChatGPT in the scale development process. Researchers tested the processes of creating and evaluating psychometric scales using ChatGPT. In this context, it was investigated how ChatGPT could play a role in item writing, determining the scale structure, and validity and reliability analyses. The study aims to reveal innovative applications of AI-supported tools in the fields of social sciences and psychometrics. The opportunities offered by AI tools can be used to accelerate scale development processes. Artificial intelligence has limitations such as producing incorrect or meaningless answers. In this study, the validity and reliability analyses of the Turkish life satisfaction scale created by ChatGPT were examined. The findings aim to contribute to both academic literature and practitioners by providing important information about the psychometric properties of artificial intelligence applications.

Method

Working group

The scale's validity and reliability were assessed using three independent samples, selected through convenience sampling. This method, characterized by its

accessibility, cost-effectiveness, and speed, facilitated data collection [22, 23]. To ensure the independence of samples, data for each analysis stage were collected from students in different faculties within the university (e.g., Engineering, Social Sciences, Medicine, etc.), with each faculty contributing data only once per stage. Additionally, a screening question (“Have you previously participated in this survey?”) was included at the start of the survey to verify that participants had not taken part in earlier stages.

The first study group is the group from which data were collected for Exploratory Factor Analysis (EFA) to be conducted to analyze the construct validity of the scale. This group consists of 503 students studying at a university in the east of Turkey in the 2024–2025 academic year. It is stated that a very good sample size is when the sample size for EFA is 500 or more [24, 25]. 297 of the students in this group are female (59%) and 206 are male (41%). The distribution of the students in this group by grades is as follows: 126 (25%) 1st year, 130 (25.9%) 2nd year, 141 (28%) 3rd year, 106 (21.1%) 4th year.

The second study group is the group from which data were collected in order to conduct Confirmatory Factor Analysis (CFA) and calculate criterion scale validity. This group consists of 301 students studying at a university in the eastern part of Turkey in the 2024–2025 academic year. It is generally accepted in the literature that data obtained from 200 to 400 participants is sufficient for CFA [26, 27]. Of the students in this group, 180 were female (60%) and 121 were male (40%). The distribution of students in this group by grades is as follows: 70 (23.2%) 1st year, 79 (26.2%) 2nd year, 68 (22.5%) 3rd year, 84 (27.9%) 4th year.

The third study group is the group in which the test–retest reliability study was conducted. This group consists of 79 students studying at a university in the eastern part of Turkey in the 2024–2025 academic year. 48 of the students in this group are female (60.7%) and 31 are male (39.3%).

Data collection tools

Artificial intelligence turkish life satisfaction scale

During the creation phase of the scale, ChatGPT-4, the paid version of OpenAI’s language model, was utilized by providing commands in the language spoken in the culture where it will be applied, “specifically Turkish”. The scale was created by giving commands in Turkish. The decision to use the paid version was made to test whether such a tool could be accessible and practical for field practitioners, such as school counselors or educators, who might rely on readily available AI tools without

extensive customization [28]. To ensure reproducibility and transparency, the specific prompt provided to ChatGPT-4 was: “Generate a 5-item Turkish life satisfaction scale in Turkish, using simple and clear language suitable for a 7-point Likert format, based on the general concept of life satisfaction” (Turkish: “Türkçe olarak, 7’li Likert formatına uygun, basit ve açık bir dille 5 maddelik bir yaşam doyumu ölçeği oluşturun”). This prompt was intentionally kept simple and broad, without providing specific scientific articles or theoretical definitions, to simulate a scenario where practitioners with limited resources might use AI tools. No pre-existing personal account with prior interactions was used; instead, a fresh session of ChatGPT-4 was initiated to minimize potential bias or influence from previous data. The scale items created by ChatGPT-4 were evaluated in terms of linguistic and cultural adaptation, clarity and understandability. In this process, three faculty members from the Guidance and Psychological Counseling and Grammar departments examined the cultural appropriateness, language structure and understandability of the items. All three of these faculty members participated in scale development studies in the field of education and psychology and conducted academic studies on life satisfaction. Faculty members evaluated the scale items written by the artificial intelligence with criteria such as compliance with the theoretical structure, clarity, comprehensibility, conciseness of expressions, and suitability for grading [29, 30]. To mitigate potential limitations, such as the generation of irrelevant or inaccurate items, this expert validation process served as a critical filter, ensuring alignment with theoretical expectations and cultural context. As a quantitative indicator of expert opinions, the Content Validity Index (CVI) of the scale was calculated. Three experts evaluated the five items of the scale using a Likert scale ranging from 1 to 4 (1 = Not at all appropriate, 4 = Completely appropriate). The Item-level Content Validity Index (I-CVI) was computed for each item by dividing the number of experts assigning a score of 3 or 4 by the total number of experts (3). The findings revealed I-CVI values of 1.00 for Items 1, 3, and 5, and 0.67 for Items 2 and 4. The Scale-level Content Validity Index (S-CVI/Ave), calculated as the average of the I-CVI values across all items, was determined to be 0.87. This value exceeds the threshold of 0.80, which is widely accepted in the literature, indicating that the scale possesses sufficient content validity [31]. However, Items 2 and 4, with I-CVI values below 0.78, drew attention in line with the experts’ suggestions for linguistic simplification and conceptual clarity. To align with the study’s objective, the scale was utilized in its original form without modification

to assess the validity and reliability of the psychometric measurement tool designed based on artificial intelligence. The scale comprises a total of 5 items and is structured in a 7-point Likert format. The Turkish version of the Artificial Intelligence Turkish Life Satisfaction Scale has been added as (Appendix 1). The similarity scan of the Life Satisfaction Scale created in ChatGPT with Turnitin (Submission Date: 15-Nov-2024 06:31OS (UTC+0300) Submission Number: 2520578326) was performed and the similarity index was found to be 0%.

Life satisfaction scale

The scale, originally developed by Diener, Emmons, Larsen, and Griffin [8] and later adapted into Turkish by Dağlı and Baysal [6], is a Likert-type instrument comprising 5 items and a single dimension. The factor loadings of the items range from 0.73 to 0.89. Confirmatory factor analysis results indicate that the obtained values fall within the acceptable ranges reported in the literature. The internal consistency of the scale, measured by Cronbach's Alpha, was found to be 0.88, while the test–retest reliability coefficient was reported as 0.97. This scale has been used in many studies before and was used in the criterion scale validity analyses of the developed scale.

General Well-Being scale short form

The scale, originally developed by Longo, Coyne, and Joseph [32] and later adapted into Turkish by [7], is a 14-item, single-dimensional likert-type instrument designed to assess individuals' affective states and overall life satisfaction. The confirmatory factor analysis conducted during the Turkish adaptation process demonstrated values within the acceptable range specified in the literature. The scale's internal consistency coefficient was determined to be 0.84, and its structural reliability was supported by a two half-test correlation of 0.77. Furthermore, the criterion-related validity analysis revealed a positive correlation between university students' general well-being, life satisfaction, positive emotions, and the fulfillment of basic needs. This scale has been used in many studies before and was used in the criterion-scale validity analyses of the developed scale.

Analysis of data

SPSS for Windows 22.0 and AMOS 21.0 package programs were used for statistical analysis of the data obtained from the scale in the study. Experts in the field were consulted for the scope validity of the scale, exploratory factor analysis was applied to the first sample group for structure validity, and confirmatory factor analysis was applied to the second sample group. Analyses related to Item-Test Correlation and Lower–Upper

Group Analysis techniques were also performed on the EFA data.

The assumptions of normality for the dataset used in Exploratory Factor Analysis (EFA) and Confirmatory Factor Analysis (CFA) were evaluated based on skewness and kurtosis values falling within the range of +1 to −1, as well as Z-scores being between +3 and −3 [25]. The analysis results indicated that the data sets utilized in both EFA and CFA adhered to the normal distribution criteria. Scale reliability was assessed using Cronbach's alpha, composite reliability (CR), and average variance extracted (AVE) coefficients derived from the EFA dataset. A Cronbach's alpha (α) value exceeding 0.7 was deemed acceptable for reliability. The discrimination power of the items was analyzed by comparing the top 27% and bottom 27% of the dataset through corrected item-total correlations [33].

In scale development and adaptation studies, factor analysis is commonly employed to assess the construct validity of a scale. This process typically involves both exploratory factor analysis (EFA) and confirmatory factor analysis (CFA) [34]. While EFA reveals the item structure and dimensions of the scale based on the conceptual basis, CFA is used to verify the structure of the scale developed with data collected from a different sample and to determine the item quality [35]. Within the scope of criterion scale validity, the “Life Satisfaction Scale” adapted to Turkish by Dağlı & Baysal [6] and the “General Well-Being Scale” adapted to Turkish by [7] were used.

Moreover, the test–retest approach was employed to assess the scale's temporal stability. In this analysis phase, Pearson correlation coefficient and intraclass correlation coefficients were calculated [36].

Findings

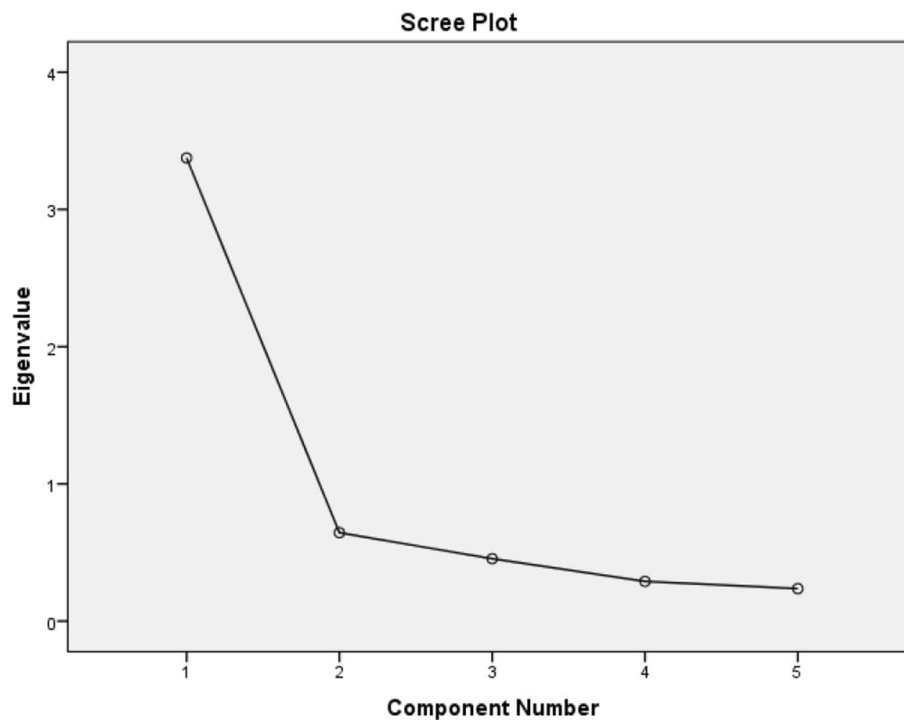
Exploratory factor analysis

To validate the Artificial Intelligence Life Satisfaction Scale, an Exploratory Factor Analysis (EFA) was conducted. The Kaiser–Meyer–Olkin (KMO) measure of sampling adequacy ($KMO=0.845$) and Bartlett's Test of Sphericity ($\chi^2=791.354$, $p<0.001$) confirmed the data's suitability for factor analysis, as the KMO value exceeded 0.60 [36]. The results of the exploratory factor analysis are shown in Table 1.

When the exploratory factor analysis (EFA) results were examined, it was determined that the factor loadings of the items ranged between 0.75 and 0.89, and the factor common variances ranged between 0.56 and 0.74. In this analysis, it was determined that the total variance explained in the scale was 67.50% and that it was a single-factor structure with an eigenvalue greater than 1. The graph in Fig. 1 also confirmed that the scale

Table 1 Exploratory factor analysis results

Items	Factor Loadings	Factor Common Variance
1. Overall, I am satisfied with my life	.859	.738
2. Most things in my life are going the way I want them to	.856	.732
3. I have healthy and happy relationships with the people I care about in my life	.781	.610
4. I think I have opportunities for self-actualization	.860	.739
5. I am optimistic about my future	.746	.557
Eigenvalue	3.38	
Total Variance Explained	67.504	
Cronbach Alpha	.88	

**Fig. 1** Turkish Life satisfaction scale factor analysis line graph

began to break after the first dimension and that the scale had a single-factor structure. Based on these findings, it can be said that the scale has a single-dimensional structure [37].

Cronbach's alpha (α) and composite reliability (CR), AVE coefficients were calculated for the scale based on EFA data. Composite Reliability (CR) and AVE were used to test the convergent validity of the scale. The results are shown in Table 2.

Table 2 AVE and CR values of artificial intelligence Turkish life satisfaction scale

Scale	Cronbach's α	CR	AVE
Artificial Intelligence Turkish Life Satisfaction Scale	.88	.90	.66

The Cronbach's alpha coefficients found for life satisfaction scale were 0.88. The CR coefficients of 0.90 for the scale. In general, reliability coefficients of 0.70 and above are presented as evidence that measurement tools can be considered reliable [38]. It is expected that the CR value of the scale is greater than the AVE value and the AVE value is greater than 0.5 [39]. In this respect, it has been determined that the CR and AVE values of the scale are at acceptable levels.

Confirmatory factor analysis

The model-data fit of the single-factor structure obtained as a result of the exploratory factor analysis was examined with confirmatory factor analysis (CFA). In the evaluation of confirmatory factor analysis results, χ^2/sd , GFI, AGFI, NFI, NNFI, IFI, CFI, RMSEA, RMR and SRMR fit indices are widely used in the literature [40, 41]. The cut-off points in Table 3 were taken as basis for these fit indices. The analysis results are given in Table 3.

When the confirmatory factor analysis results in Table 3 are examined, $\chi^2/sd=2.63$, GFI=0.99, AGFI=0.95, NFI=0.99, NNFI/TLI=0.98, IFI=0.99, CFI=0.99, RMSEA=0.07, RMR=0.04 and SRMR=0.03 are obtained. When the fit index values of the scale are examined based on these results, it is seen in Table 3 that the values obtained have perfect fit or acceptable fit values. The model obtained as a result of CFA is shown in Fig. 2 below.

Overall, it is observed that factor loadings in the CFA are generally above 0.50. However, it is noteworthy that the loading of one item falls below 0.50. Research suggests that CFA factor loadings should be at least 0.3, though this value may vary depending on the context. For a stronger relationship, factor loadings of 0.45 and above are preferred, as they indicate a more meaningful association between the variable and the factor [44]. Both exploratory and confirmatory factor analyses, along with

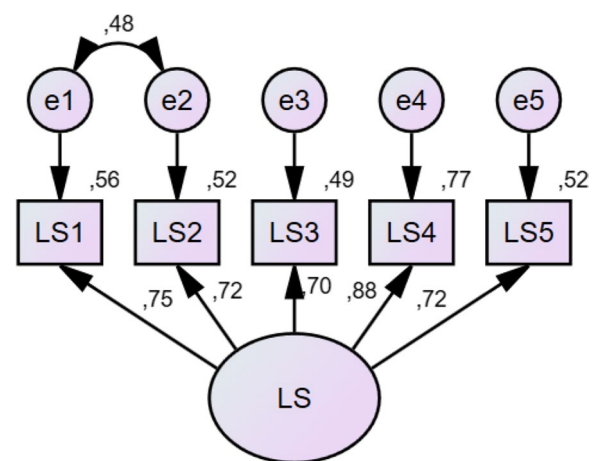


Fig. 2 Confirmatory factor analysis final model

reliability and validity analyses and item analyses, demonstrate that the scale possesses a consistent structure. Additionally, life satisfaction scales used in the literature are also observed to exhibit a unidimensional structure [8, 45, 46]. Criterion scale validity was calculated using the Life Satisfaction Scale adapted to Turkish by Dağlı & Baysal [6] and the General Well-Being Scale Short Form adapted to Turkish by Odacı et al. [7] based on the data on which confirmatory factor analysis was performed.

When Table 4 is examined, a positive relationship ($r=0.74$) was observed between the Artificial Intelligence Turkish Life satisfaction scale and the Dağlı & Baysal [6] Life satisfaction scale, and a positive relationship ($r=0.63$) was observed between the ChatGPT Life satisfaction scale and the [7] General well-being scale. In addition, a positive relationship ($r=0.68$) was observed between the Dağlı & Baysal [6] Life satisfaction scale and the [7] General well-being scale.

Cronbach Alpha reliability coefficient was calculated on the CFA data of the scale and found to be .89. In addition, the scale was applied to 79 students at 18-day intervals in order to determine whether the scale was consistent over time. The arithmetic mean, standard deviation, Cronbach Alpha coefficients and test-retest correlation coefficients obtained in both applications are given in Table 4.

Table 3 Acceptable and good fit values for fit indices

Fit Index	Acceptable Fit	Perfect Fit	Value Obtained from CFA
χ^2/sd	$2 \leq \chi^2/sd \leq 5$	$0 \leq \chi^2/sd < 2$	2.63 (Acceptable Fit)
GFI	$0.90 \leq GFI < 0.95$	$0.95 \leq GFI \leq 1.00$	0.99 (Perfect Fit)
AGFI	$0.85 \leq AGFI < 0.90$	$0.90 \leq AGFI \leq 1.00$	0.95 (Perfect Fit)
NFI	$0.90 \leq NFI < 0.95$	$0.95 \leq NFI \leq 1.00$	0.99 (Perfect Fit)
NNFI/TLI	$0.95 \leq NNFI < 0.97$	$0.97 \leq NNFI \leq 1.00$	0.98 (Perfect Fit)
IFI	$0.90 \leq IFI < 0.95$	$0.95 \leq IFI \leq 1.00$	0.99 (Perfect Fit)
CFI	$0.95 \leq CFI < 0.97$	$0.97 \leq CFI \leq 1.00$	0.99 (Perfect Fit)
RMSEA	$0.05 \leq RMSEA \leq 0.08$	$0 \leq RMSEA < 0.05$	0.07 (Acceptable Fit)
RMR	$0.05 \leq RMR \leq 0.08$	$0 \leq RMR < 0.05$	0.04 (Perfect Fit)
SRMR	$0.05 \leq SRMR \leq 0.08$	$0 \leq SRMR < 0.05$	0.03 (Perfect Fit)

[41–43]

Table 4 Correlation coefficients relating to criterion scale validity

Scales	1	2	3
1. Artificial Intelligence Turkish Life Satisfaction	-	-	
2. Dağlı & Baysal [6] Life Satisfaction	.74**	-	
3. Odacı etc. [7] General Well-being	.63**	.68**	-

** $p < .01$

Table 5 Test–retest reliability analysis results

Application	$\bar{X}$	r	Cronbach Alpha
I. Application	3.85	.95	.89
II. Application	3.84		.88

According to the results obtained in Table 5, the arithmetic mean of the scores in the first application was 3.85, the Cronbach Alpha internal consistency coefficient was 0.89. In the second application, the arithmetic mean of the scores was 3.84, the Cronbach Alpha internal consistency coefficient was 0.88. The correlation coefficient calculated to determine the consistency of the scale over time was found to be “0.95”. There is a positive and highly significant relationship between the pre-test and post-test measurements of the scale. In addition, Intraclass correlation coefficients (ICC) were calculated in the test–retest analysis and are shown in Table 6. According to these results, it can be said that the scale is consistent over time.

Item analysis

The item analyses of the scale were performed using the Item-Test Correlation methods and Sub-Upper Group Analysis techniques on the EFA data [36]. The t-test was used to determine whether there was a significant difference between the upper group 27% and the lower group

27% of the life satisfaction scale. The results of the item analyses of the scale are given in Table 7.

Using a 27% cutoff value to define lower and upper participant groups, the t-test results revealed that the t-values for differences between these groups ranged from 4.25 to 6.28. All items demonstrated statistically significant t-values, indicating a meaningful distinction between the lower 27% and upper 27% groups. These significant t-values provide evidence supporting the discriminatory power of the items [36]. The corrected item-total correlations presented in Table 7 indicate that all items exceed 0.40. This suggests that each item makes a significant contribution to the total score and supports the scale’s strong discriminative power [47, 48]. The composite reliability (0.90) and average variance extracted (0.66) further substantiate the scale’s internal consistency and convergent validity. These values demonstrate that, despite its brevity, the scale exhibits robust psychometric properties [49, 50]. Table 7 also shows that item-total correlations range from 0.62 to 0.77. When the results are evaluated collectively, it is concluded that each item of the Artificial Intelligence Turkish Life Satisfaction Scale is discriminative. Before conducting measurement invariance analyses, the model-data fit indices for the original factor structure of the Artificial Intelligence Turkish Life Satisfaction Scale according to gender are presented in Table 8.

Upon examining Table 8, it is observed that the fit indices of the AI Turkish Life Satisfaction Scale for

Table 6 Intraclass correlation coefficient (ICC) of the artificial intelligence Turkish life satisfaction scale

Items	Intraclass correlation coefficient (ICC)
1. Overall, I am satisfied with my life	0.91
2. Most things in my life are going the way I want them to	0.93
3. I have healthy and happy relationships with the people I care about in my life	0.92
4. I think I have opportunities for self-actualization	0.90
5. I am optimistic about my future	0.93

Table 7 Results of item analysis of artificial intelligence Turkish life satisfaction scale

Items	Corrected Item Total Correlations	Upper (%27) X	Lower (%27) X	Lower -Upper 27% t-Test	p
1. Overall, I am satisfied with my life	.752	4.26	1.62	4.25	<.001
2. Most things in my life are going the way I want them to	.749	4.39	1.63	5.36	<.001
3. I have healthy and happy relationships with the people I care about in my life	.653	3.99	1.38	5.72	<.001
4. I think I have opportunities for self-actualization	.772	4.23	1.66	6.28	<.001
5. I am optimistic about my future	.617	4.02	1.52	5.82	<.001

Table 8 Fit indices of the artificial intelligence Turkish life satisfaction scale for subgroups

Groups	χ^2	df	RMSEA (%90 CI)	SRMR	CFI	TLI
Women	15.342	5	0.070 (0.035–0.105)	0.035	0.95	0.93
Men	14.875	5	0.071 (0.034–0.108)	0.038	0.94	0.92

Note: df = degrees of freedom, confidence intervals for RMSEA values (90%) are given in parentheses

the gender variable fall within the ranges commonly accepted in the literature for assessing model-data fit ($RMSEA < 0.08$, $CFI > 0.90$, $TLI > 0.90$, $SRMR < 0.08$). In this context, the findings provide evidence that the single-factor structure of the AI Turkish Life Satisfaction Scale is consistent with the data obtained from all subgroups. In other words, the original factor structure is confirmed in each subgroup, and construct validity is established across all subgroups.

Findings regarding measurement invariance

Measurement invariance of the AI Turkish Life Satisfaction Scale across gender was assessed using multi-group confirmatory factor analysis (MG-CFA), testing configural, metric, and scalar invariance models. Results are presented in Table 9.

The configural invariance model showed acceptable fit ($RMSEA = 0.070$, $CFI = 0.945$, $SRMR = 0.036$), indicating that the scale's single-factor structure is consistent across gender groups. Metric invariance, with equal factor loadings across groups, was supported by acceptable fit indices ($RMSEA = 0.068$, $CFI = 0.940$, $SRMR = 0.039$) and small changes in fit ($\Delta CFI = 0.005$, $\Delta RMSEA = 0.002$), suggesting that the scale's items measure the same construct equivalently for females and males. Scalar invariance, with equal item intercepts, also showed adequate fit ($RMSEA = 0.071$, $CFI = 0.925$, $SRMR = 0.042$; $\Delta CFI = 0.005$, $\Delta RMSEA = 0.003$), confirming that mean comparisons across genders are valid. These results demonstrate that the AI Turkish Life Satisfaction Scale exhibits measurement invariance at configural, metric, and scalar levels across gender, supporting its use for gender-based comparisons [51, 52].

Discussion, conclusion and recommendations

This study investigates the psychometric properties of the Artificial Intelligence Turkish Life Satisfaction Scale developed using ChatGPT, contributing valuable insights into the validity and reliability of AI applications in psychometrics. These analyses offer significant contributions to academic literature and practical applications, while also providing guidance for future scale development efforts. The research focused on the validity and reliability analyses of the Artificial Intelligence Turkish Life Satisfaction Scale developed by ChatGPT in order to measure the life satisfaction levels of university students. In conclusion, this research has demonstrated that artificial intelligence-based tools can be evaluated and trusted in terms of psychometrics by revealing the validity and reliability properties of the Artificial Intelligence Turkish Life Satisfaction Scale, and has also provided valuable findings that will shed light on future scale development studies.

As a result of the analyses, it was determined that the scale had a single-factor structure, the factor loadings of the scale items varied between 0.75 and 0.86, the factor common variances were between 0.56 and 0.74, and 5 items explained approximately 68% of the total variance. It was determined that the corrected item test correlation coefficients of the scale were between 0.62 and 0.77. As a result of the CFA performed on the scale, which was determined to be one-dimensional, it was determined that the scale's goodness of fit values were perfect or acceptable when the values in the literature were taken into account [41–43]. The Cronbach alpha internal consistency coefficient of the scale was calculated as 0.88. It was determined that the CR and AVE values calculated for the scale were within the limits accepted in the literature. According to the test–retest analysis results of the developed scale, it can be said that it is stable against time. Within the scope of the item analyses of the scale, it was found that there was no difference between the

Table 9 Multi-group CFA results for the artificial intelligence Turkish life satisfaction scale

Variable	χ^2	df	RMSEA	CFI	SRMR	$\Delta\chi^2$	Δsd	p	ΔCFI	$\Delta RMSEA$	$\Delta SRMR$
Gender											
Configural	30.217	10	0.070	0.945	0.036						
Metric	35.482	14	0.068	0.940	0.039	5.265	4	0.26	0.005	0.002	0.003
Scalar	50.673	18	0.071	0.925	0.042	15.191	4	0.40	0.005	0.003	0.003

upper group 27% and the lower group 27%. As a result of the analyses, it was concluded that the developed scale has a one-dimensional structure, its psychometric properties are within the acceptable limits specified in the literature, it has high internal consistency and validity values and can be evaluated as a stable measurement tool against time.

The exploratory factor analysis of the Life Satisfaction Scale, adapted into Turkish by Dağlı and Baysal [6], indicated that the factor loadings of the scale items ranged between 0.73 and 0.89. The confirmatory factor analysis results fell within the acceptable range specified in the literature. Additionally, the Cronbach's alpha coefficient for internal consistency was calculated as 0.88. These findings from the validity and reliability analyses demonstrate that the Turkish version of the Life Satisfaction Scale, as adapted by Dağlı and Baysal [6], aligns with the standards in the literature and serves as a reliable measurement instrument.

The General Well-Being Scale Short Form, adapted into Turkish by Odacı et al. [7], is a unidimensional scale comprising 14 items. The confirmatory factor analysis results indicate that the obtained values fall within the acceptable range established in the literature. The scale demonstrated a Cronbach's alpha internal consistency coefficient of 0.84, reflecting a high level of reliability. These findings align closely with the results obtained from the Artificial Intelligence Life Satisfaction Scale. Overall, the validity and reliability outcomes of the Turkish adaptation suggest that the General Well-Being Scale Short Form is a psychometrically sound and dependable instrument for measurement.

Findings have been obtained that the Artificial Intelligence Turkish Life Satisfaction Scale developed with ChatGPT support has criterion scale validity. Criterion scale validity is the correlation value obtained with data collected simultaneously from the same group with a scale that previously measured the same or similar feature of a developed scale [53]. When the correlation value is medium and above in criterion scale validity, it is reported in studies as evidence of the criterion scale validity of the developed scale [54]. These findings reveal that the Artificial Intelligence Turkish Life Satisfaction Scale developed with ChatGPT support has criterion scale validity by showing a significant and sufficient relationship with scales measuring similar features.

In the study, a positive relationship ($r=0.74$) was found between the Artificial Intelligence Turkish Life Satisfaction scale and the Dağlı & Baysal [6] Life satisfaction scale. A positive relationship ($r=0.63$) was determined between the Artificial Intelligence Life Satisfaction scale and the [7] General well-being scale. Within the scope of the Turkish adaptation of the [7] Life Satisfaction scale, the criterion scale validity was calculated with the Life Satisfaction Scale

adapted to Turkish by [55]. The researchers obtained a positive significant relationship (0.61) between these two scales. In this context, it was observed that the findings of the scale developed with artificial intelligence support and the scales developed by field experts gave similar results. In addition, a positive relationship ($r=0.68$) was observed between the Dağlı & Baysal [19] Life satisfaction scale and the [7] General well-being scale in this study. The research findings reveal that there are high-level positive relationships between the Artificial Intelligence Turkish Life Satisfaction Scale and the scales developed by field experts, and that these scales give parallel and consistent results.

ChatGPT's learning ability indicates the potential to develop meaningful and reliable scales without human intervention. In this study, it was observed that the Artificial Intelligence Turkish Life Satisfaction Scale created by ChatGPT yielded similar results when compared to scales developed by field experts. The research findings show that a valid and reliable scale was obtained to measure the life satisfaction of university students. It is anticipated that this research will provide an important resource for studies to be conducted on the life satisfaction of university students in Turkey. In addition, it is thought that the validity and reliability analyses conducted with the life satisfaction scale created by ChatGPT will make significant contributions both in the academic field and in the application processes and will pave the way for the development of future psychometric scales.

Considering the productivity potential of ChatGPT, a conversational AI chatbot developed and supervised by OpenAI, which operates using a large language model optimized through reinforcement learning methods [56, 57], it is anticipated to contribute to scale development processes. A limitation of AI is its reliance on existing data, which may compromise its originality. Artificial intelligence models, particularly those based on deep learning algorithms, are trained on vast amounts of data. This training process enables the model to learn patterns and relationships within the data. However, this also introduces the risk that the content generated by AI may not be entirely original. AI can reorganize, combine, or rephrase information present in the training data. Consequently, the content produced by AI may be regarded as a derivative of existing data rather than an original creation. Additionally, the lack of critical thinking in AI may constitute another limitation [58, 59].

The training of artificial intelligence (AI) models, particularly deep learning algorithms, on large datasets significantly enhances their abilities in knowledge acquisition and pattern recognition. However, this raises questions about the originality of the content generated by AI. Specifically, AI can restructure, combine, or rephrase existing

information from its training data. Consequently, the content produced by AI may be considered a derivative of existing data rather than a wholly new creation. This situation opens up a discussion regarding the potential and limitations of AI, particularly in fields requiring creativity.

In addition, the limited critical thinking capacity of AI represents a significant constraint. Humans, when analyzing information, evaluate different perspectives, reconcile contradictory evidence, and generate novel ideas. In contrast, AI models typically produce outcomes based on existing data and may struggle to address complex, abstract, or ambiguous situations. This limitation makes it challenging for AI to replace human expertise, particularly in complex decision-making processes and creative problem-solving domains [58–60]. Therefore, artificial intelligence should be utilized with care and precision in scale development processes.

One notable methodological challenge in this study pertains to the design of the prompt used for AI-driven item generation. The prompt provided to ChatGPT-4 was intentionally kept simple and broad—“Generate a 5-item life satisfaction scale in Turkish, using simple and clear language suitable for a 7-point Likert format, based on the general concept of life satisfaction”—to avoid overly directing the AI and to simulate a scenario where practitioners with limited resources might use AI tools. While this approach ensured that the AI's output was not unduly constrained, it also posed the risk of producing generic or less contextually nuanced items. For instance, items 2 and 4, which received lower Item-level Content Validity Index (I-CVI) scores (0.67), were flagged by experts for needing linguistic simplification and greater conceptual clarity. This highlights the delicate balance between crafting non-leading prompts and mitigating the risk of overly general outputs. Designing effective prompts for AI-driven scale development requires careful consideration of specificity, cultural relevance, and theoretical grounding to ensure that generated items are both psychometrically robust and contextually appropriate. Future research could explore strategies for optimizing prompt design, such as incorporating domain-specific keywords, iterative prompt refinement, or hybrid approaches that combine AI-generated items with expert-driven modifications to enhance item quality and relevance.

Despite the promising results, this study has several methodological and conceptual limitations that warrant critical examination. First, the study's reliance on a convenience sample from a single university in eastern Turkey limits the generalizability of the findings. The sample, primarily composed of young adult university students, may not adequately represent diverse populations with varying age groups, socioeconomic backgrounds, or cultural contexts, potentially restricting the scale's applicability across broader demographics. Second, the use of a broad and simple prompt for ChatGPT-4, while practical for simulating

real-world applications, resulted in items that were occasionally generic, as evidenced by the lower I-CVI scores for items 2 and 4. This suggests that AI-generated items may lack the nuanced specificity required to fully capture complex constructs like life satisfaction, particularly in culturally specific contexts. Third, the test–retest analysis was conducted over a relatively short interval (18 days), which may not adequately capture the scale's stability over longer periods, especially given that life satisfaction can fluctuate due to external events such as economic or social changes. Fourth, while the scale demonstrated measurement invariance across gender, its invariance across other demographic variables, such as age or socioeconomic status, was not tested, limiting the understanding of its robustness across diverse groups. Finally, the exclusive validation of the scale in Turkish raises questions about its adaptability to other languages and cultural settings, as cross-cultural validation was beyond the scope of this study.

To address these limitations, future research could adopt several strategies. First, replicating the study with more diverse and representative samples, including participants from different regions, age groups, and socioeconomic backgrounds, would enhance the scale's generalizability. Second, refining AI prompt design by incorporating more specific theoretical frameworks or cultural contextual cues could improve the quality and relevance of generated items. For example, prompts could include references to established life satisfaction theories (e.g., [21]) or culturally salient factors. Third, conducting test–retest analyses over extended time intervals (e.g., 1, 3, or 6 months) would provide stronger evidence of the scale's temporal stability. Fourth, testing measurement invariance across additional demographic variables, such as age, education level, or cultural background, would further validate the scale's applicability across diverse populations. Finally, translating and validating the scale in other languages and cultural contexts would facilitate its use in cross-cultural research, contributing to a more global understanding of life satisfaction.

In conclusion, this study demonstrates that AI-supported tools, such as ChatGPT, can produce psychometrically robust scales, as evidenced by the validity and reliability of the Artificial Intelligence Turkish Life Satisfaction Scale. These findings highlight the transformative potential of AI in streamlining scale development processes while maintaining scientific rigor. However, the identified limitations underscore the need for cautious and critical application of AI in psychometrics, emphasizing the importance of human oversight and robust validation. By addressing these challenges, future research can further harness AI's capabilities to advance psychological measurement, offering innovative tools to explore complex constructs like life satisfaction across diverse populations.

Appendix I

Artificial Intelligence Turkish Life Satisfaction Scale (Yapay Zeka Türkçe Yaşam Doyumu Ölçeği)	Kesinlikle katılmıyorum	Katılmıyorum	Kısmen katılmıyorum	Ne katılıyorum ne katılmıyorum	Kısmen katılıyorum	Katılıyorum	Kesinlikle katılıyorum
1. Genel olarak, yaşamımdan memnunum	1	2	3	4	5	6	7
2. Hayatımdaki çoğu şey istediğim gibi gidiyor	1	2	3	4	5	6	7
3. Hayatımda değer verdiğim insanlarla sağlıklı ve mutlu ilişkilerim var	1	2	3	4	5	6	7
4. Kendimi gerçekleştirme fırsatlarına sahip olduğumu düşünüyorum	1	2	3	4	5	6	7
5. Geleceğim konusunda iyimserim	1	2	3	4	5	6	7

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Artificial intelligence usage disclosure

The authors confirm that no private or individually identifiable data were disclosed to AI during the item generation process for this article. They acknowledge that ethical concerns regarding privacy and confidentiality become particularly prominent when personal data are fed into AI systems for the purposes of research, authorship, and/or publication of this article. All data used were processed in accordance with relevant privacy standards and ethical guidelines. The authors used ChatGPT to improve the language quality of the article during the preparation of this study.

Authors' contributions

Authors contributed equally to the article. All authors reviewed the manuscript.

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Data availability

Our dataset named Validity and Reliability Analysis of the Life Satisfaction Scale Developed with Artificial Intelligence (view at <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/L3KAA2>) was published in Harvard Dataverse (view at <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/L3KAA2>).

Declarations

Ethics approval and consent to participate

All procedures performed in studies involving human participants comply with the ethical standards of the institutional ethics committee and the 1964 Declaration of Helsinki and its subsequent amendments or comparable ethical standards. The study was reviewed and approved by Inonu University Social and Human Sciences Ethics Committee. Ethics Committee Decision No: 2024/18–7.

Informed consent

All individual participants involved in the study provided their informed consent.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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